

Anderson acceleration for PWR whole-core pin-by-pin Nu-TH coupling calculation in NECP-Bamboo2.0

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Paper Overview

- NECP-Bamboo2.0 is a full-cycle simulation tool
 - Designed to simulate full cores
 - Includes neutronics, TH (via START), chemistry, control system, depletion
 - Developed by Xi'an Jiaotong University
- Details the implementation of Anderson Acceleration (AA)
 - Improvement to typically used Picard iteration
 - Implementation verified against VERA problems 6 and 7

Multi-Physics Phenomenon

Neutronics

- Whole-core, pin-by-pin neutronics calculations
 - Typical codes use perform assembly homogenized calculations
- Two neutronic sub-modules
 - Lattice solves 2D transport to obtain XS for high-fidelity calculations
 - Core takes Lattice's XS and solves SP_3 for each pin
 - SP_3 is the simplified diffusion method using third order Legendre
- Obtains transport-like accuracy with diffusion-like speeds

Thermal Hydraulics (TH)

- Steady and Transient Analysis for Reactor Thermal-hydraulics (START)
 - Directly coupled to the neutronics module
 - Separate code developed by the same group as Bamboo (NECP)
- START is a two-phase, homogeneous flow, subchannel, fluid solver
 - Also handles heat transfer in the multiphysics solution

Coupling

- Feedback between neutronics and thermal-hydraulics is tightly coupled
- Proper feedback requires two-way coupling
- Used to use Picard iteration
 - Typical coupling method
 - Eventually the computation bottleneck as runtimes were too long

Computational Methods

Picard Iteration

- Multi-physics coupling technique that involves...
 1. Solving the first physics field
 2. Passing the first-field solution as an initial condition to the second field
 3. Solving the second physics field
 4. Passing the second-field solution as an initial condition to the first field
 5. Solving the first field and checking the residual between old/new solns
 6. Iterating until both physics fields converge

- Modified with the use of relaxation factors
 - New solution is a combination of previous and current solutions

Anderson Acceleration (AA)

- Improvement to Picard iteration
 - Residual minimization technique
 - Similar to the generalized minimal residual (GMRES) method
- Solution represented as combination of multiple previous solutions
 - Optimal weighted average of past solutions that minimized residual
 - Less overshooting than Picard iteration
 - More stable than Picard iteration

Problem Statements

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VERA Benchmarks

- VERA Core Physics Benchmark Problems 6 and 7
 - Series of benchmarks based on the Watts Bar Nuclear 1 reactor
 - PWR Designed by Westinghouse
- Problem 6: solving for Nu-TH feedback on 17x17 assembly
- Problem 7: whole-core criticality search at hot, full-power

Path of Accomplishment

1. Identify Picard iteration is the main bottleneck
2. Implement AA, which does not require much code refactoring
3. Incrementally increase test problems
 - a. Single pin-cell for basic feedback
 - b. 2D assembly for radial feedback
 - c. VERA problem 6
 - d. VERA problem 7
4. Benchmarking against existing codes

Problems

1. Strong feedback between fuel temperature and water density
2. How many previous solutions should be used in AA?
3. Partial convergence interference between Nu and TH solvers

Solutions

1. If the solutions oscillates/residual grows, revert to previous soln
2. Perform study using various solution depth
3. Synchronize the convergence criteria to ensure convergence before global iterations

Results

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VERA Problem 6

- Comparison against Picard iteration
 - Best Picard iteration took 222 global iterations and 294 seconds
 - Best AA took 6 global iterations and 11.3 seconds
- AA far outperformed Picard iteration
- Benchmark against MPACT/CTF
 - Very close criticality agreement of +24pcm
 - Predicted max pin-power distribution bias of -0.48%
- Excellent agreement with reference solution
- Performed similarly to MC21/CTF and MC21/COBRA-1E

VERA Problem 7

- Comparison against Picard iteration
 - Best Picard iteration took 14 global iterations and 176.9 seconds
 - Best AA took 8 global iterations and 111.4 seconds
- Nu-AA outperformed Picard iteration, but TH-AA did not
- Benchmark against MPACT/CTF
 - Very close Critical boron concentration agreement of -4pcm
 - Max assembly power distribution bias of 0.69%
- Excellent agreement with reference solution
- Performed similarly to MC21/CTF and MC21/COBRA-1E

Conclusions

- Implementation of Anderson acceleration in NECP-Bamboo2.0 was a success
- Notable performance increase over Picard iteration
- Extremely close agreement with VERA benchmark problems 6 and 7
 - Compared to MPACT/CTF
- Results similar to existing, established codes
 - MC21/CTF and MC21/COBRA-1E